Lightning Storm End Prediction

DataBattle 2026

Team: SolveIt

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Outline

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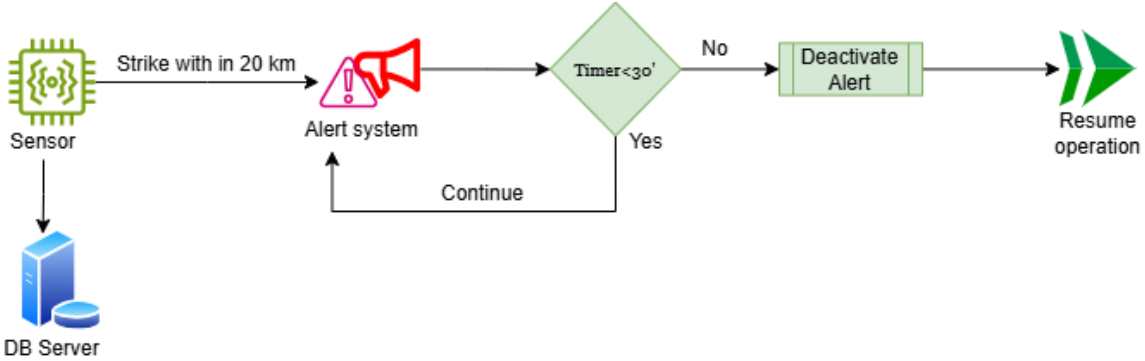
Overview

Current Operational Rule

- Airport aprons are shut whenever a CG strike occurs within a **20 km alert zone**.
- Strikes in the **20–50 km** outer ring provide early-warning context.
- The all-clear requires **30 minutes of silence** since the last CG strike.



Problem Statement

Key Issues

- Fixed 30–60 min rule after last strike $\rightarrow$ not adaptive.
- Causes delays or early decisions in critical operations.
- Ignores spatiotemporal lightning patterns for accurate prediction.

Our Goals

- Develop a real-time probabilistic model to estimate storm end likelihood
- Analyze storm patterns at each airport
- Improve decision-making and reduce unnecessary alert durations

Proposed Solution

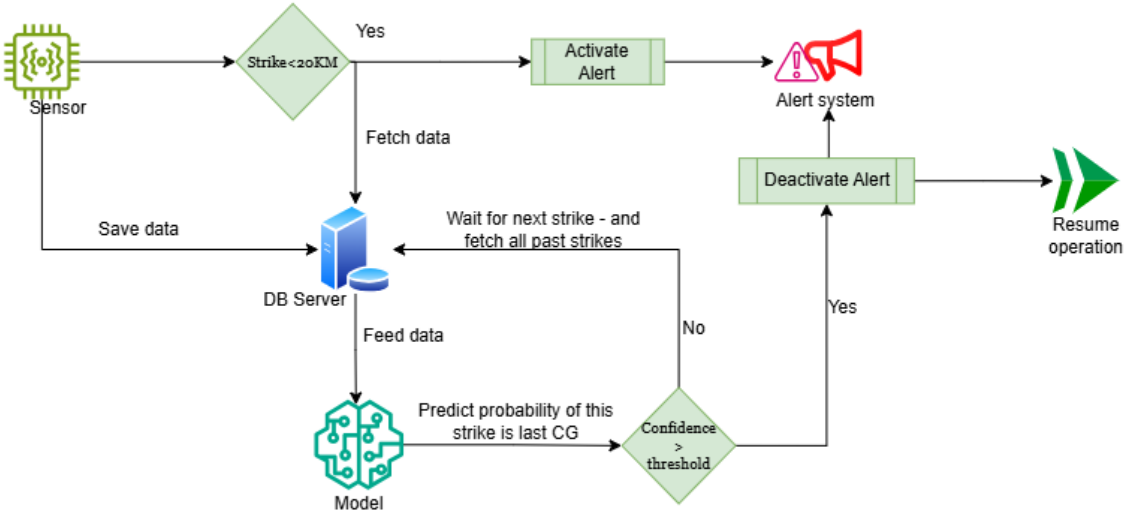


Figure: Proposed Solution Architecture

Data Overview

Dataset at a glance

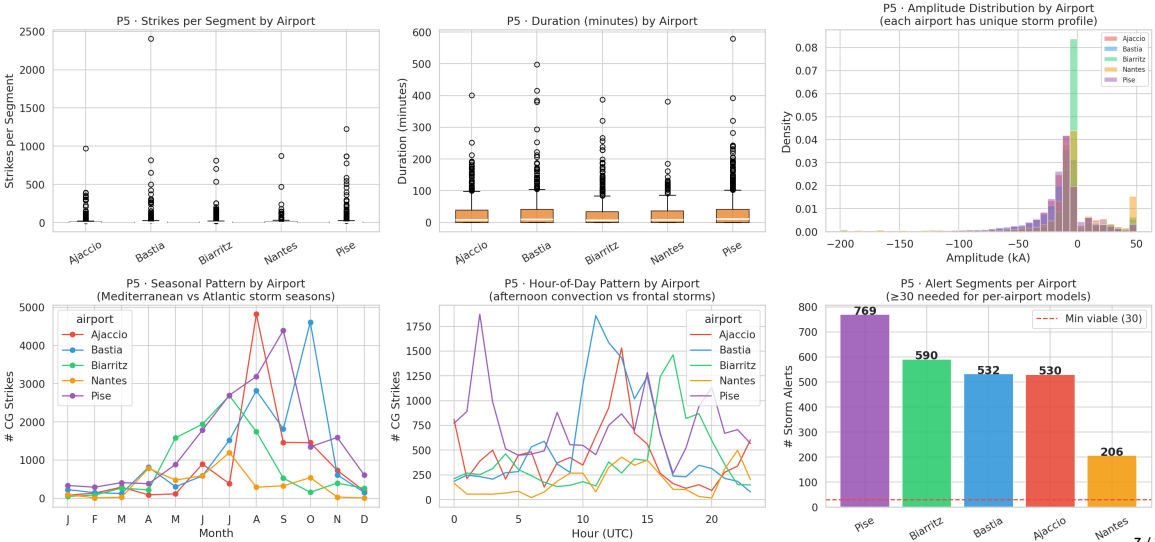
Total strikes (raw)	**507 071**
Labelled CG strikes	**56 599**
Positive class (last CG)	**2 627 (4.6%)**
Airports	**5 French airports**
Years	**2016–2022**

Data preprocessing

- Missing data: alert_id, is_last_cg_strike b/se they are cloud strikes
- No duplicates
- Outliers: Are physically valid Meteorage sensor readings, and removing these extreme values would eliminate the most informative late-stage storm signals.
- Class imbalance: is_last_cg_strike has 1:21 (True:False) class ratio and addressed
  - class weighting (20×), stratified GroupKFold, and OOF-based threshold tuning (optimal threshold = 0.71)

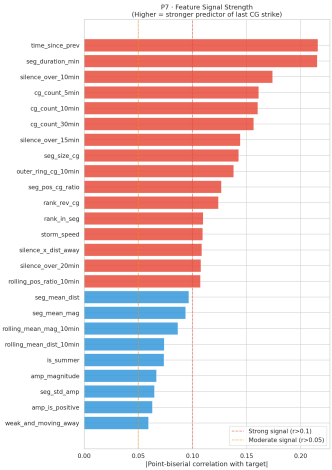
EDA — Storm Patterns

PART 5 · Airport Profiles



Feature Engineering — 36 Features across 11 Groups

Group	N	Key signal
A – Amplitude	3	Polarity, magnitude
B – Segment ctx	6	Segment statistics
C – Position	3	Rank, `pct_position`
D – Temporal lag	4	Time gap, spatial drift
E – Rolling	5	10/30-min activity
F – Silence	3	Silence >10/15/20 min
G – Interaction	2	Weak & moving away
H – Outer ring	1	CG outer 10-min ring
I – Calendar	4	Hour, month, season
J – Raw physics	5	Dist, azimuth, x/y
K – Airport enc	1	Target encoding
Total	**36**	



Point-biserial correlation ranking. `pct_position` and silence features lead.

Modeling & Evaluation

- Our task: For each cloud-to-ground (CG) strike, predict the **probability** it is the **last strike** before 30 min of silence:

Models

- Logistic regression
- Light gradient boosting machine
- eXtreme Gradient Boosting

Metrics

- Area Under Curve(AUC)
- F-1 Score test
- Brier (probability calibration)

GroupKFold(k=5)

- Groups = `segment_key` (airport + alert ID)
- No segment spans train *and* validation
- All features computed within-segment, in strict temporal order
- Verified with explicit leakage audit

Temporal stress-test

Train $\leq$ 2020 / Validate $\geq$ 2021
AUC holds $\Rightarrow$ model generalises across years

Three Models — Same Folds, Same Features

Model	AUC	F1	Brier
Logistic Regression	0.9430	0.352	0.1064
XGBoost	0.9800	0.774	0.0164
LightGBM	**0.9810**	**0.797**	**0.0140**
Blind baseline	—	—	*0.047*

LightGBM wins on every metric

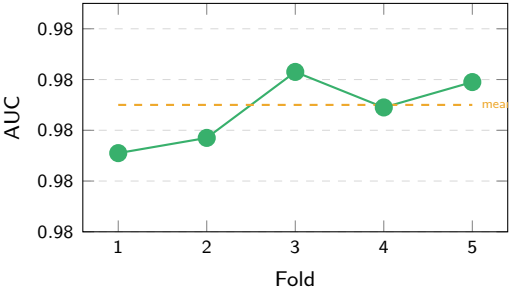
- Brier score −34% vs blind baseline
- F1 +2.3 pp vs XGBoost
- AUC std ≈ 0.001 — very low variance

What we achieved

- Brier 0.014 vs baseline 0.044 (−34%)
- **FPR 0.15%** — 81 false all-clears out of 53,972 active strikes: model is **extremely safe**
- FNR 29.2% — safe failure mode: short extra wait, never a premature all-clear
- Near-perfect calibration, consistent across 5 airports
- Measured CO_2: **0.34 g total training**

LightGBM — Cross-Validation Detail

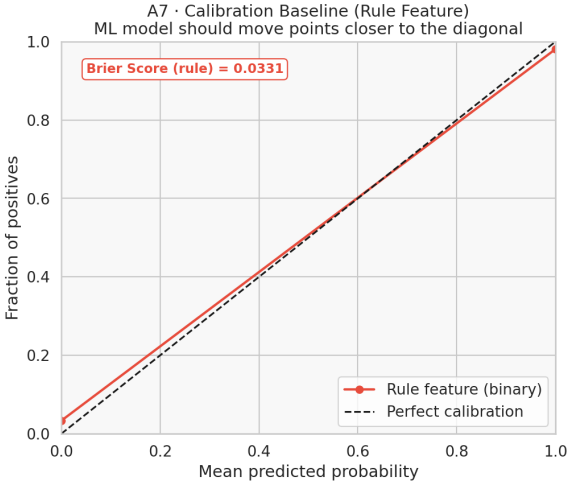
Fold	AUC	F1	Brier
1	0.9791	0.7943	0.01407
2	0.9797	0.8000	0.01418
3	0.9823	0.8100	0.01350
4	0.9809	0.7963	0.01418
5	0.9819	0.7826	0.01418
Mean	**0.9810**	**0.7966**	**0.01402**



Decision threshold: **0.71** (tuned on OOF to maximise F1)

Low variance confirms the model generalises consistently across Mediterranean and Atlantic airports.

Probability Calibration



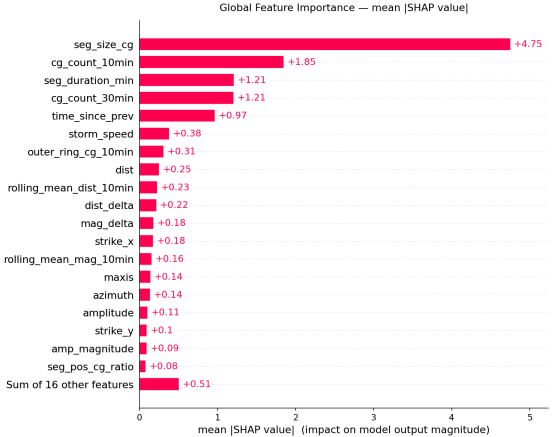
Why calibration matters

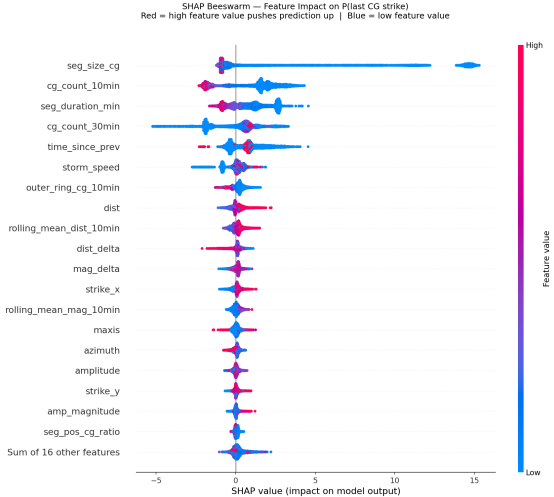
- $\hat{p} = 0.80$ means an 80% real probability
- ATC can set any threshold matching their own safety / delay trade-off
- Enables accountable, auditable decisions

Result

LightGBM reliability curve closely follows the diagonal — near-perfect calibration *without* post-hoc recalibration. Threshold tuned to **0.71** on OOF predictions.

SHAP — Global Feature Importance





pct_position **silence_over_20min** **rank_rev_cg** are the top-3 drivers. Storm position dominates over raw physics.

Direct Environmental Cost (Measured)

Life-Cycle Assessment — measured (CodeCarbon)

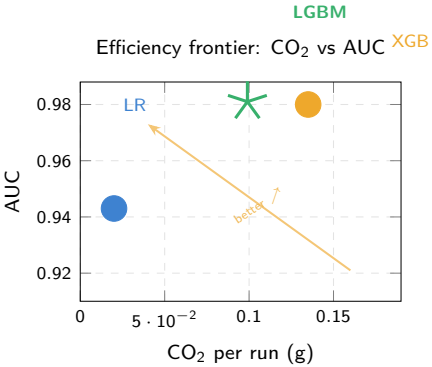
- **Training** (5-fold CV, 56 K samples): **0.34 g** CO_2, **1.04 Wh**, **2.6 min**
- **Inference** (56 K CG strikes, full dataset): **0.015 g** CO_2, **0.065 Wh**
- Per alert (single storm): $<$**0.003 g** CO_2
- Equivalent to charging a phone for **12 min** (training only)

Hardware & infrastructure

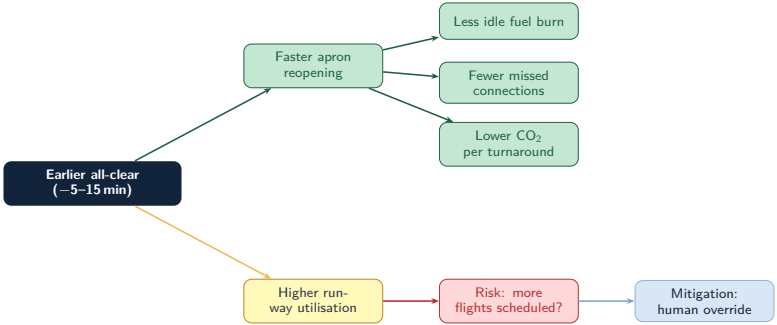
- **CPU-only** — no GPU procured or rented
- No new hardware purchased
- Self-contained Docker image — no cloud API, no data-centre allocation
- Water footprint: negligible (no cooling tower)

Sustainability shaped model choice

LightGBM: **0.099 g** CO_2/run XGBoost: 0.135 g (+36%) LR: 0.020 g but F1 = 0.35. **Best accuracy AND lower CO_2** — no trade-off needed.



Indirect Effects and Consequence Tree



Legend

- Benefit
- Indirect
- Risk
- Mitigation

Following best practices: multi-criteria analysis, rebound effects identified, mitigation proposed (Rasoldier 2022; Ekchajzer 2024).

Conclusion

Real-Time Prediction

Probabilistic model replaces the fixed 30-min rule with **adaptive, calibrated** decision-support — enabling earlier, safer all-clears.

Operational Value

Faster apron reopening, fewer delays, less idle fuel burn, and lower CO_2 per turnaround across 100+ French airports.

Proven Performance

LightGBM: AUC **0.981** ▪ F1 **0.797** ▪ Brier **0.014**
FPR **0.15%** — safe and consistent across all 5 airports.

Green & Explainable AI

Training: **0.34 g CO_2**. Inference per storm alert: $<$**0.003 g** — both measured via CodeCarbon. SHAP transparency ensures accountable decisions.

Adaptive storm-end prediction: safer, greener, and operationally ready.

Thank You

Questions?

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